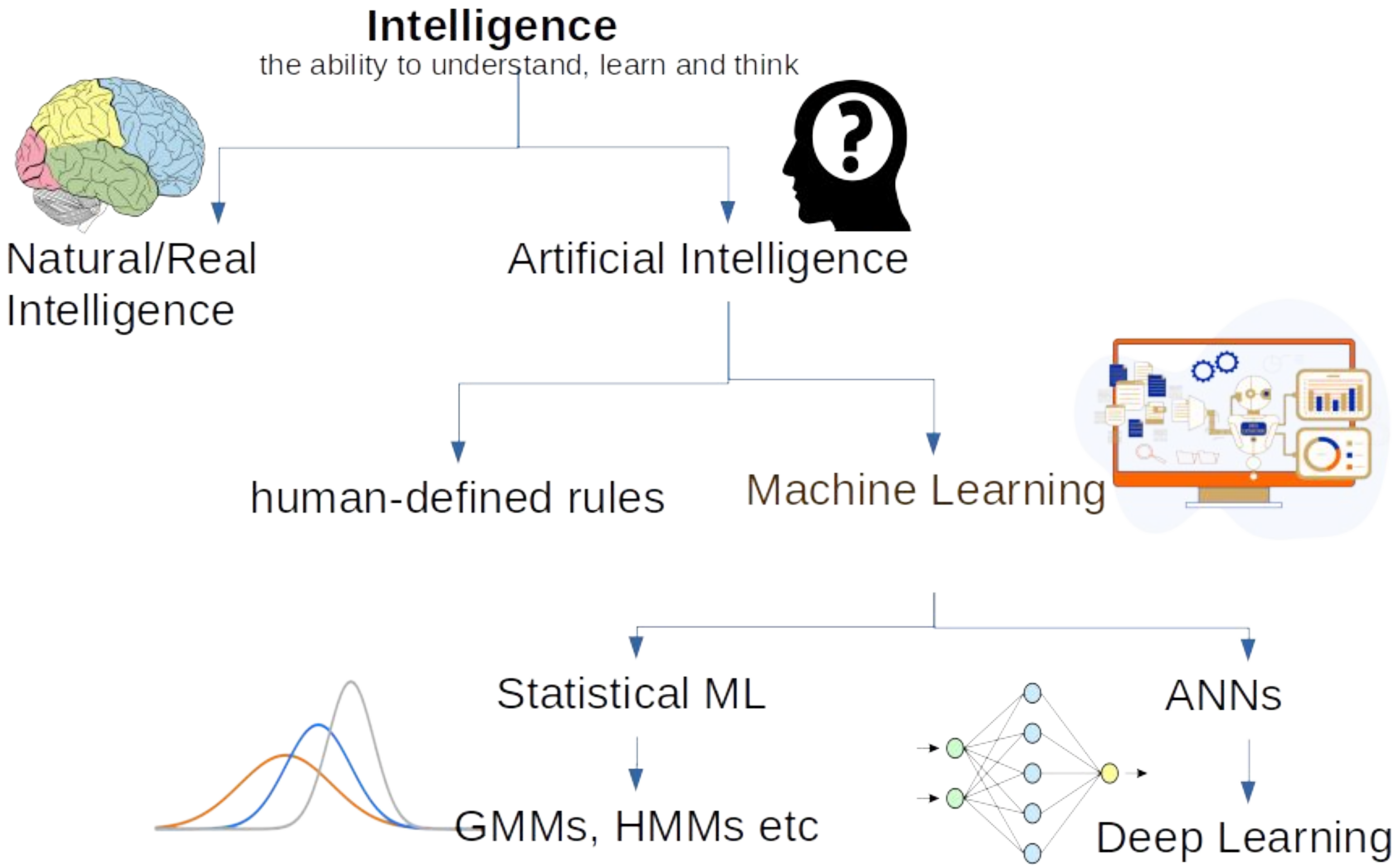
Deep Learning

Review
UG-4, Aug 2025-26
Shaik Mohammad Rafi B



Intelligence
the ability to understand, learn and think
Natural/Real Intelligence
Artificial Intelligence
human-defined rules
Machine Learning
Statistical ML
GMMs, HMMs etc
ANNs
Deep Learning

Raw data

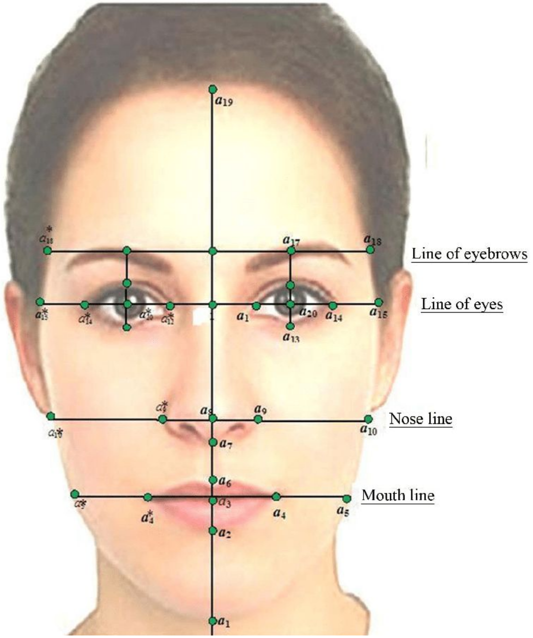
Documents → **1Dim:** Articles, records, stories, poems

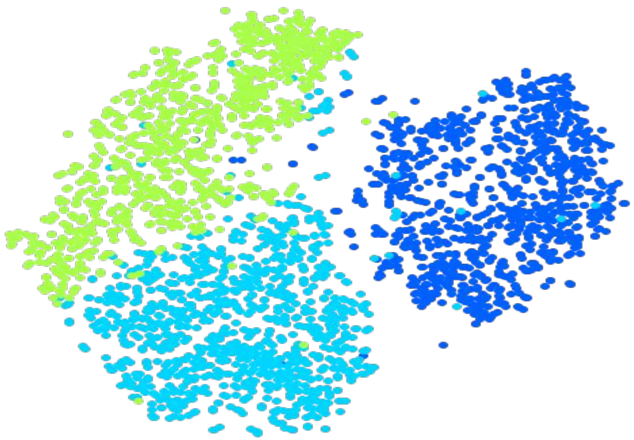
Images → **2Dim:** Photographs, satellite, thermal, medical

Numbers → **1Dim:** EEG, ECG, Seismic, weather, stock market

Sounds → **1Dim:** Speech, Music, bird/animal, Vehicles etc

Features & patterns

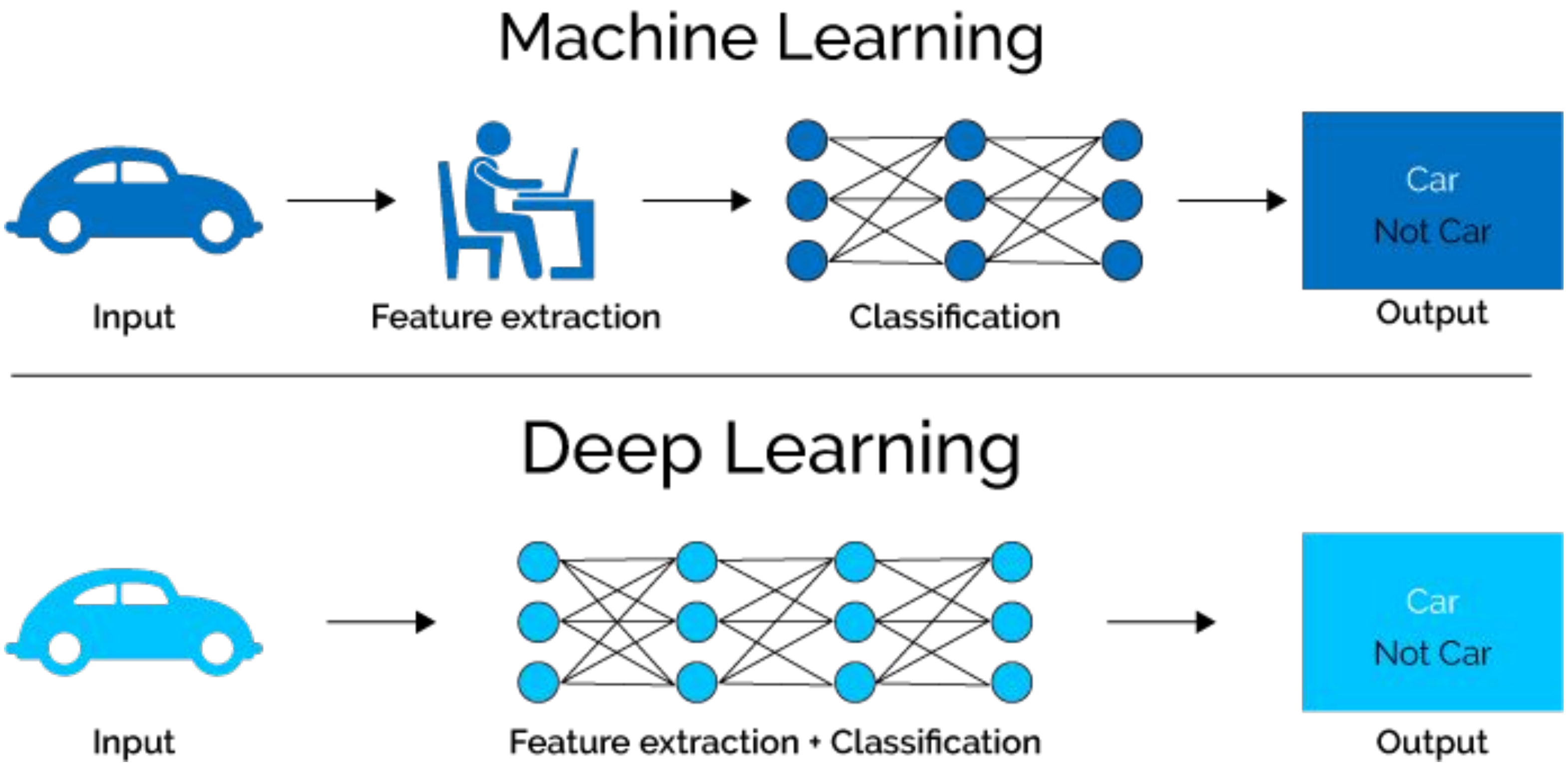




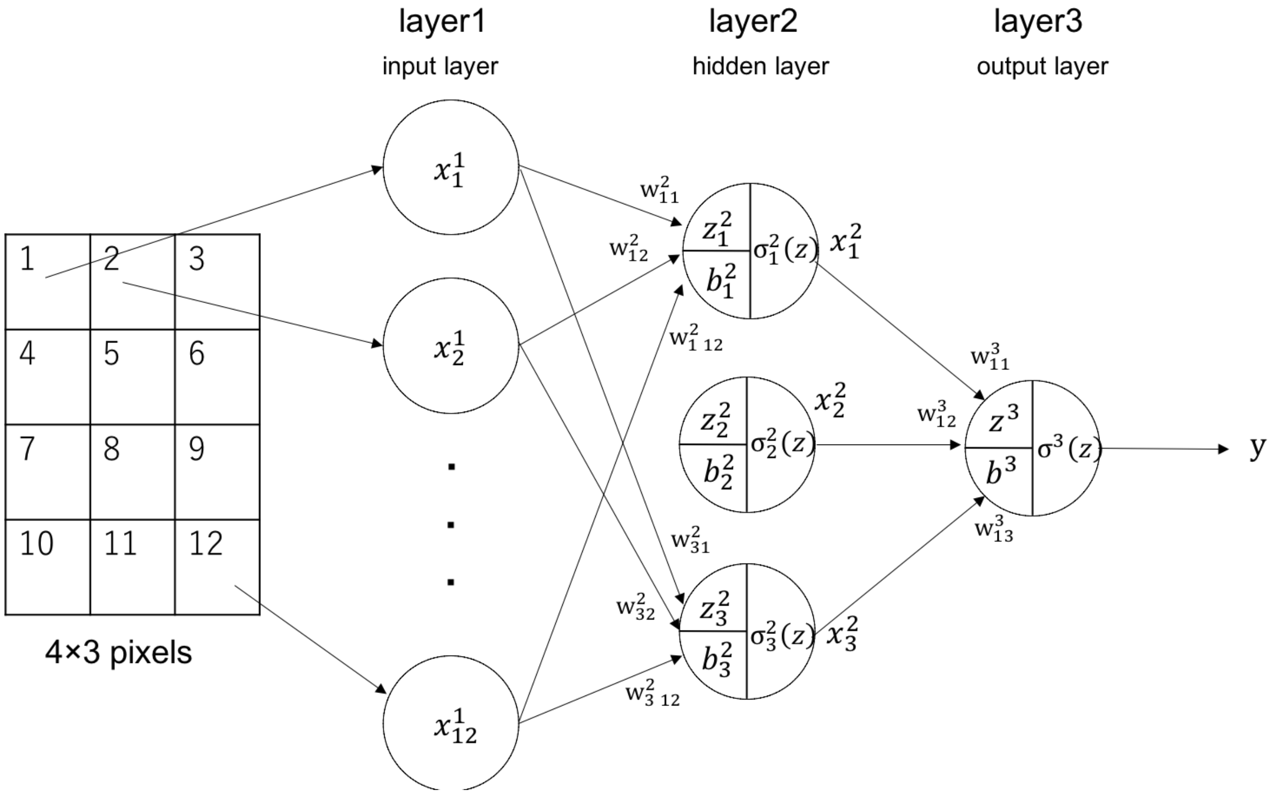
Features:
Measurable descriptors of an entity

Pattern:
Systematic arrangement or combination of features

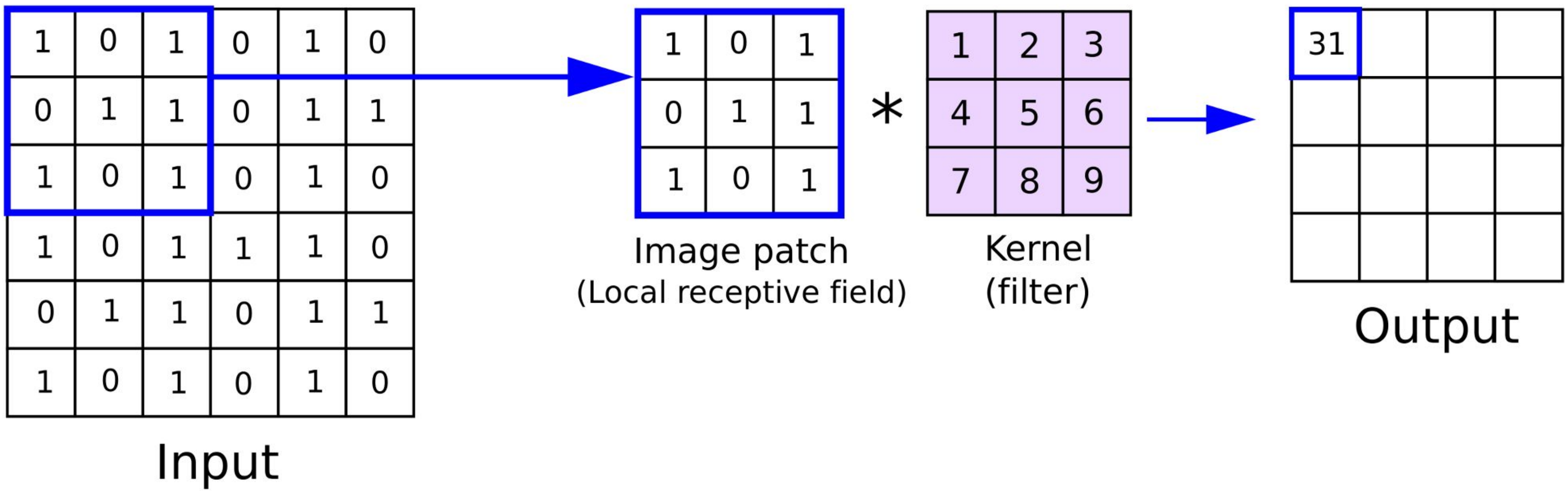
Feature learning through Deep Neural Nets



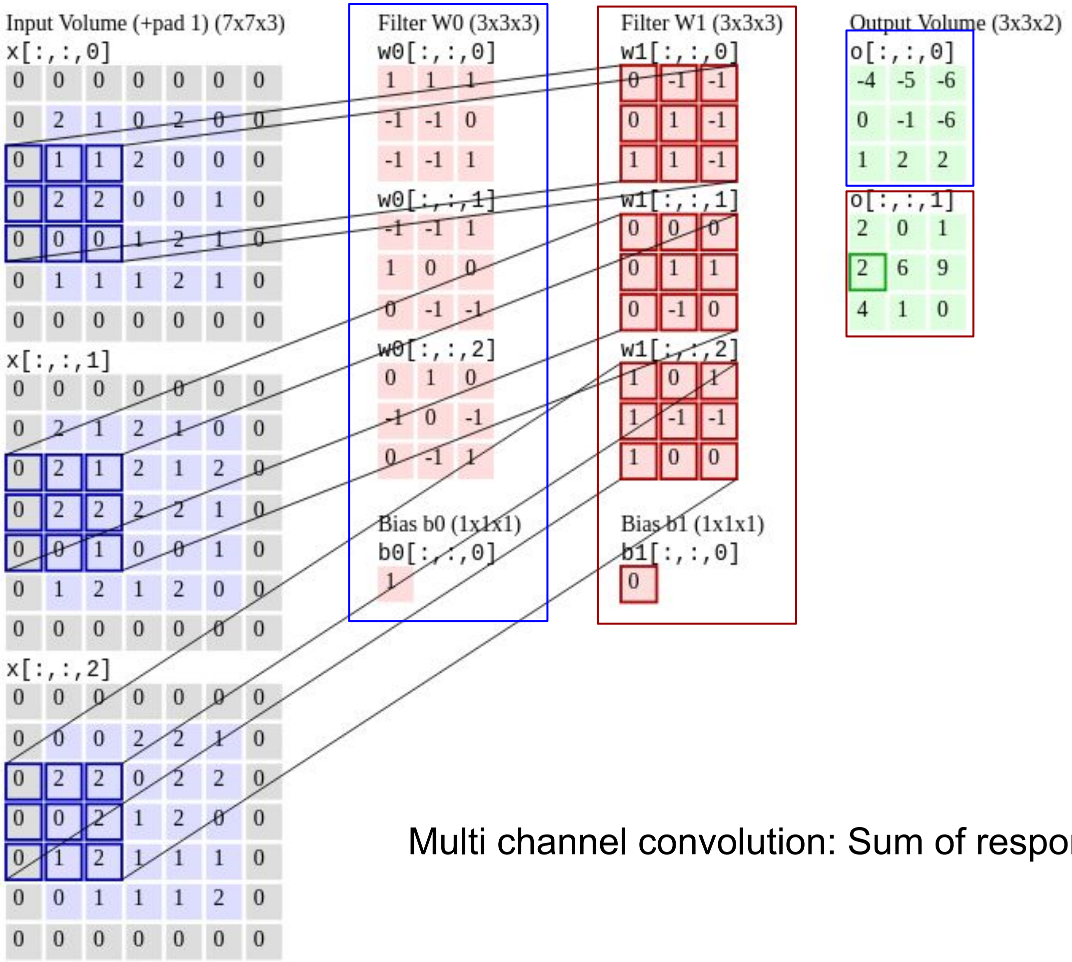
Feed Forward Neural Network



Convolutional Neural network

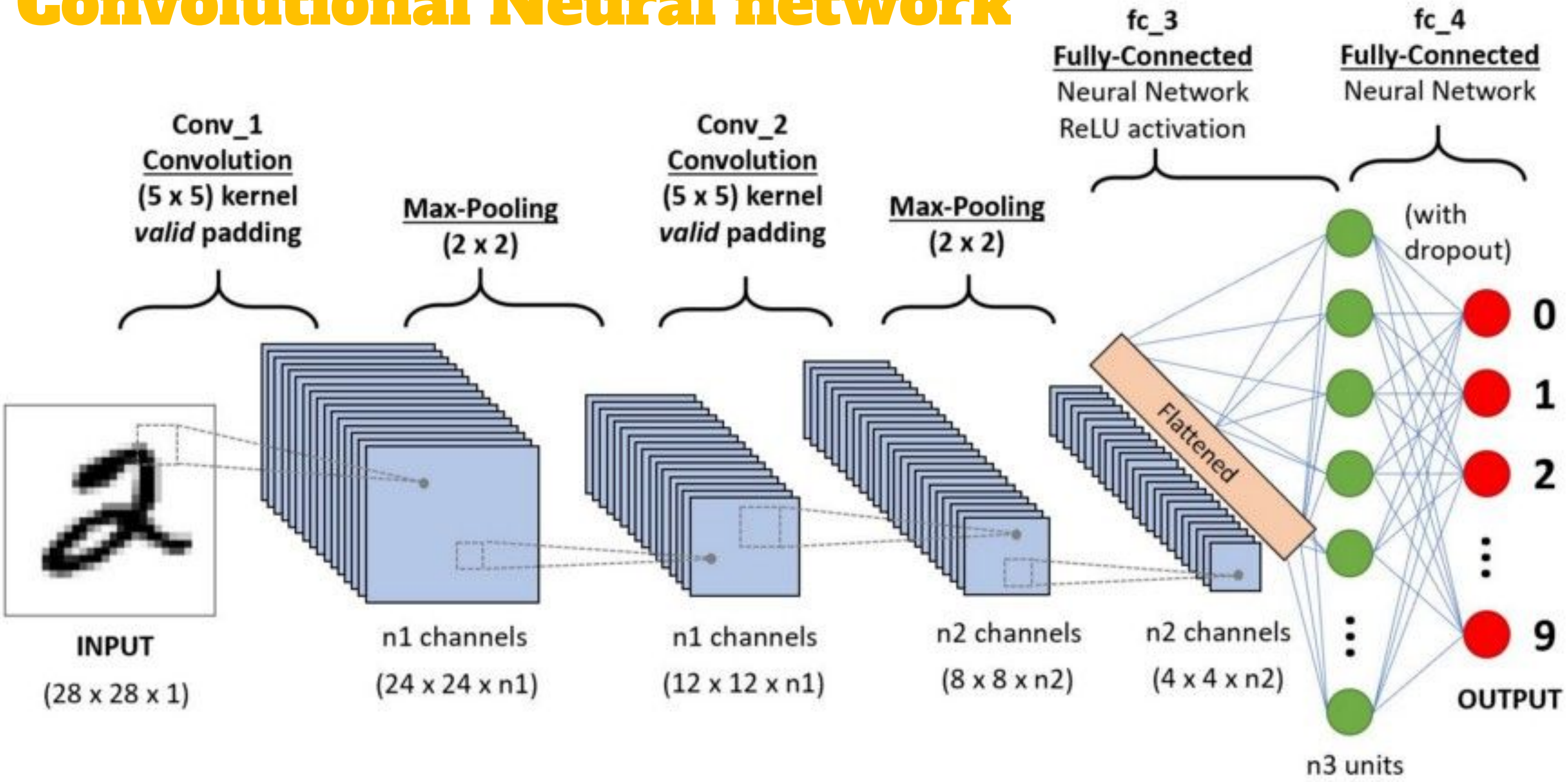


Convolution Operation

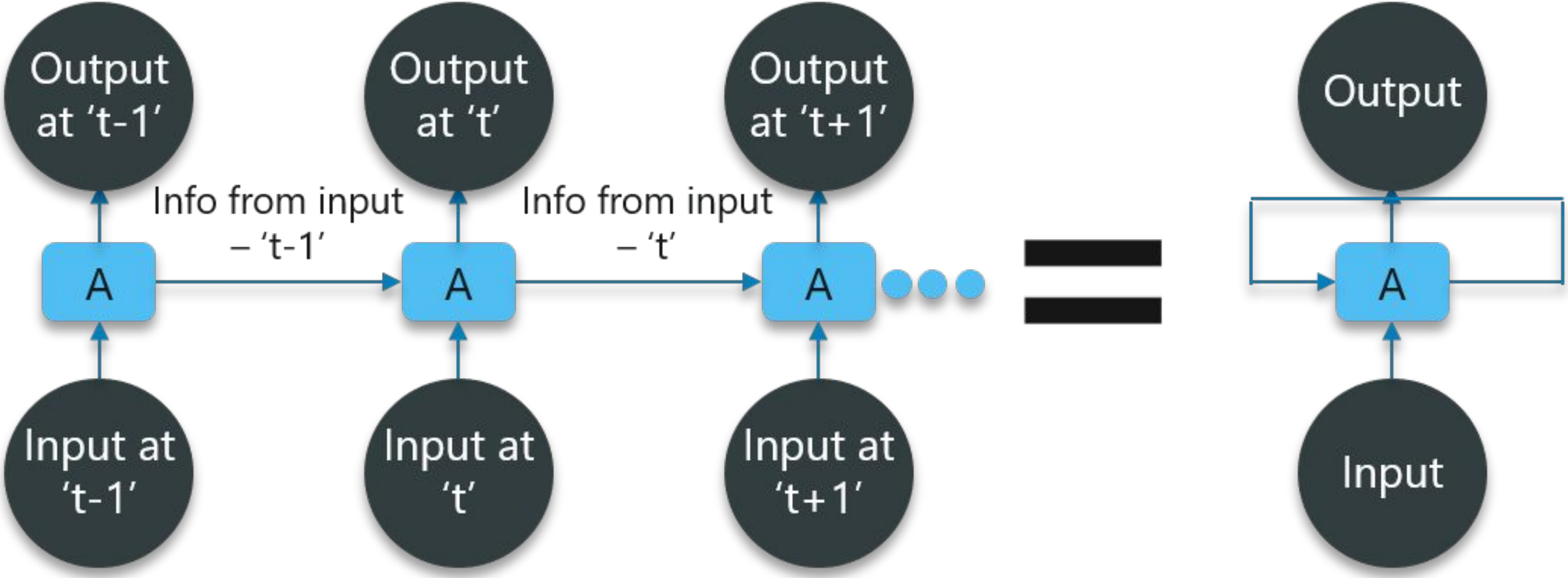


Multi channel convolution: Sum of responses

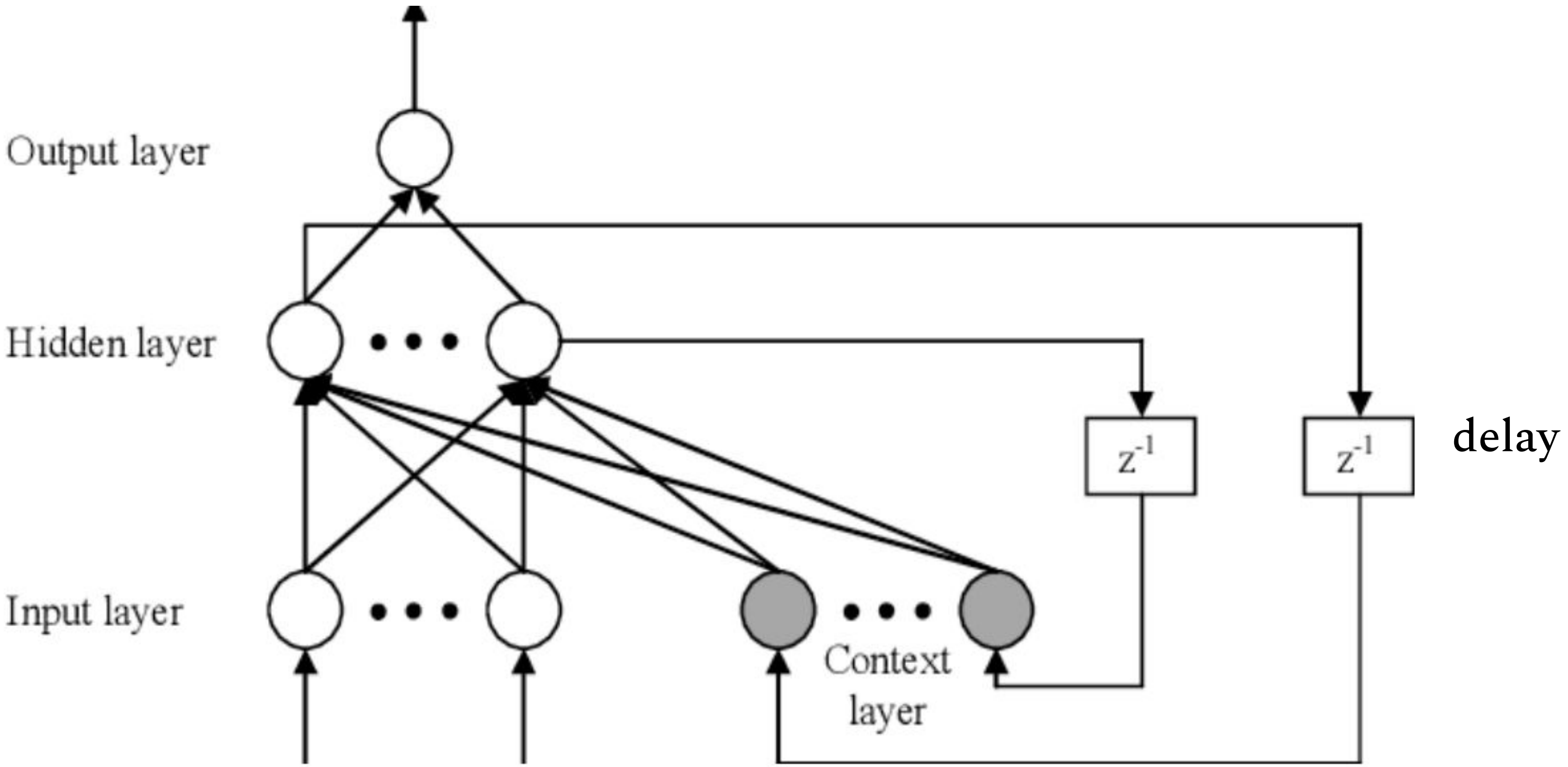
Convolutional Neural network



Recurrent Neural Network

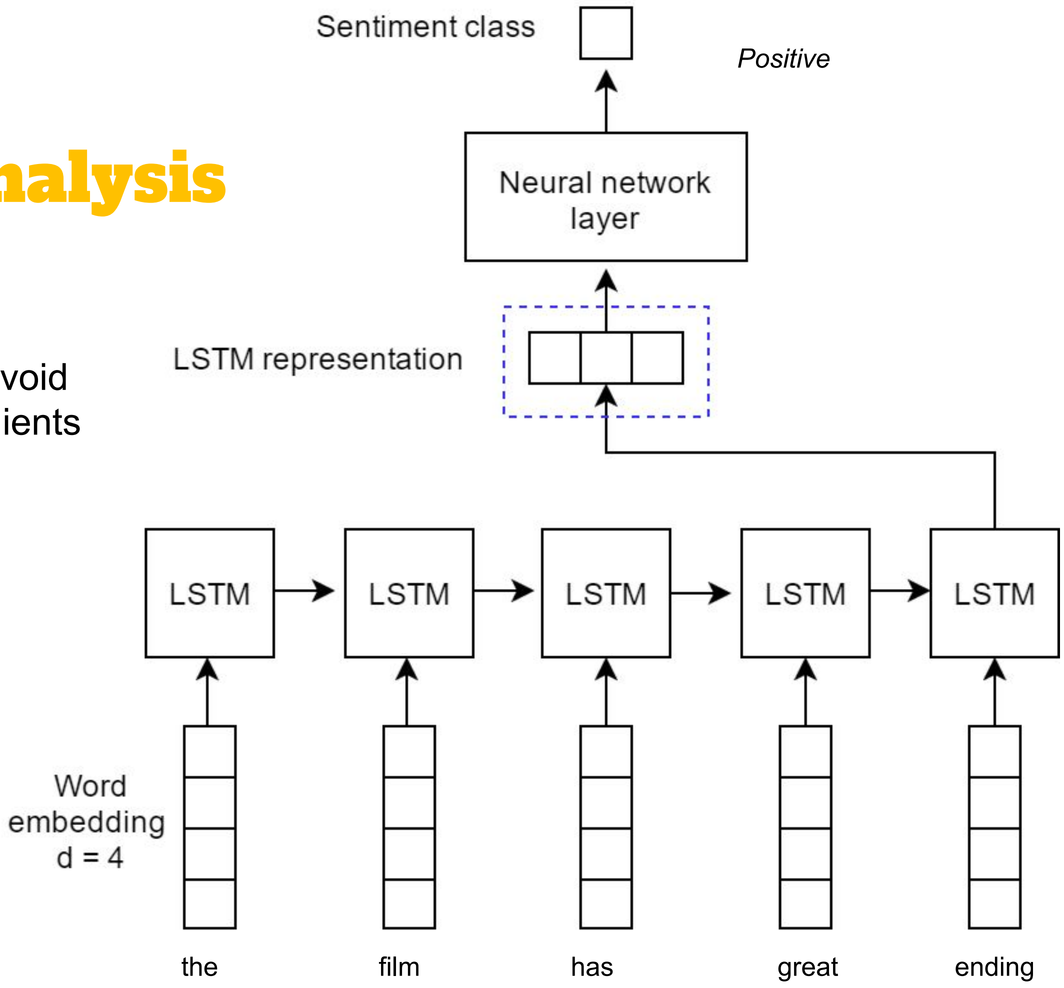


Recurrent Neural Network

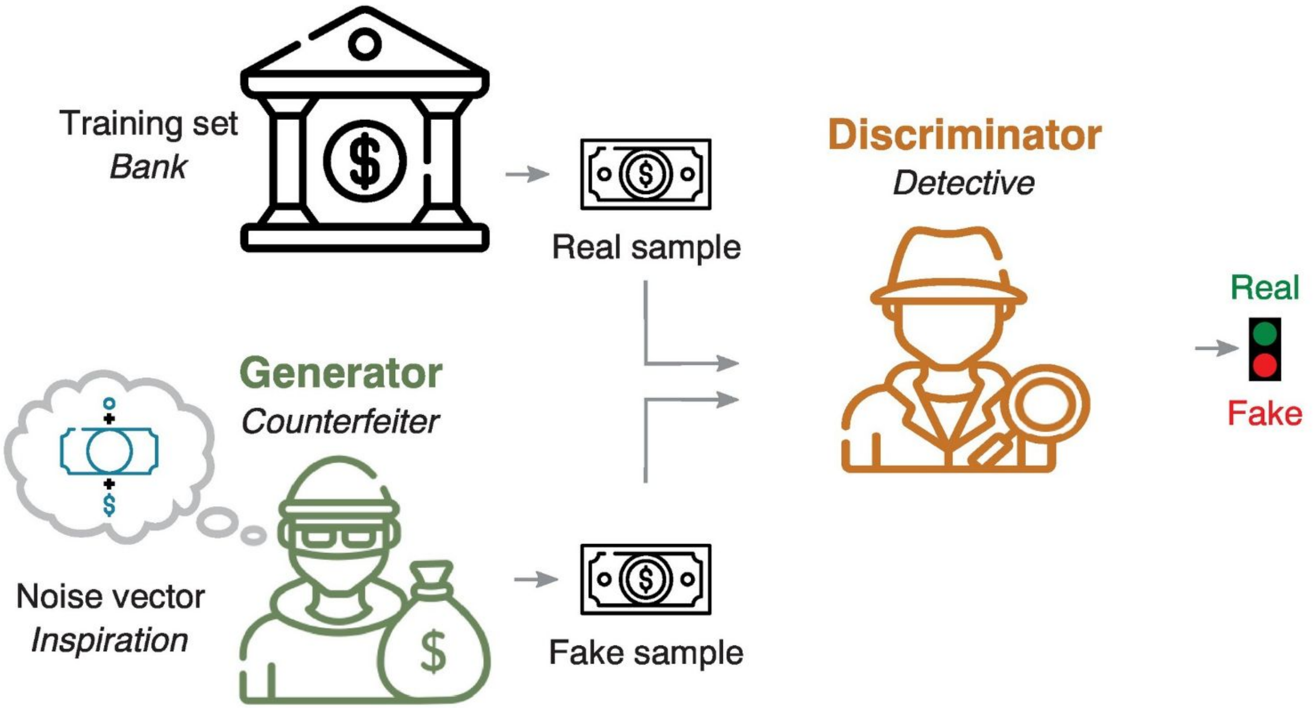


Sentiment Analysis

LSTM replaces RNN cell to avoid vanishing and exploding gradients through gating mechanism.



Generative Adversarial Network



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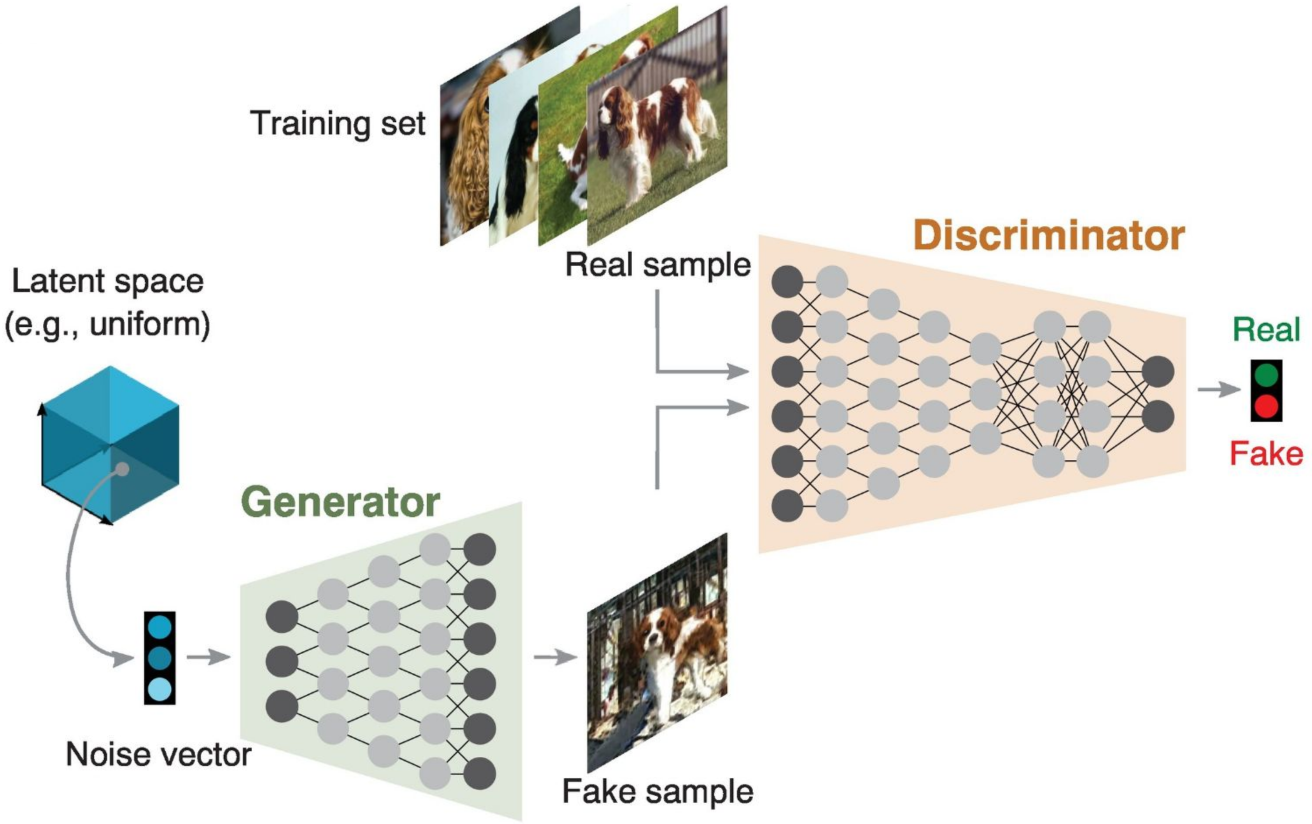
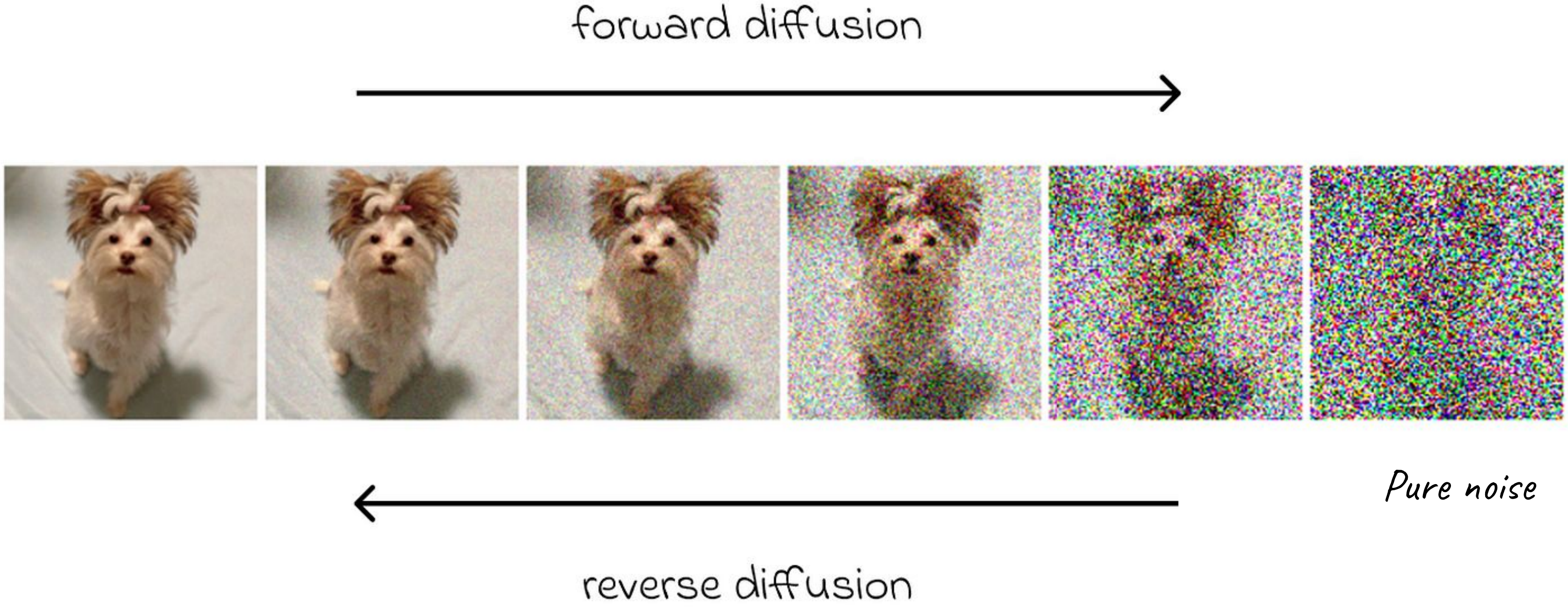
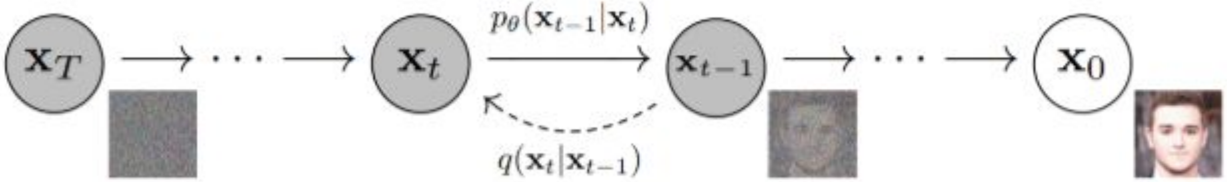


Diffusion Models

Deep Generative models: Generate real data from noise



- The goal of the model is to detect the added noise and reconstruct the previous image.
- The predicted image is then compared to the actual image to calculate the loss.

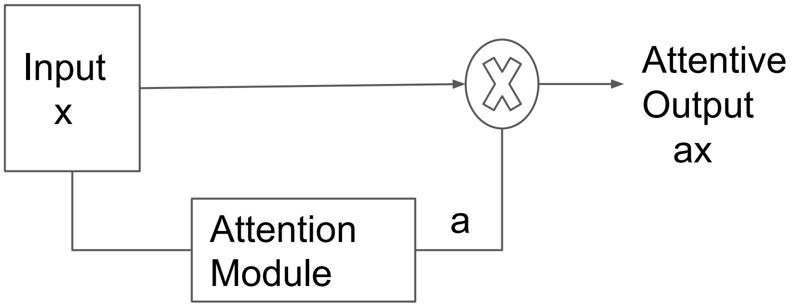


- The noisy image at step t and the value of t itself is input to the model.
- The model predicts the noise that was added to the image to get it to that state.
- Mean Squared Error (MSE) between the *actual* noise added and the *predicted* noise.

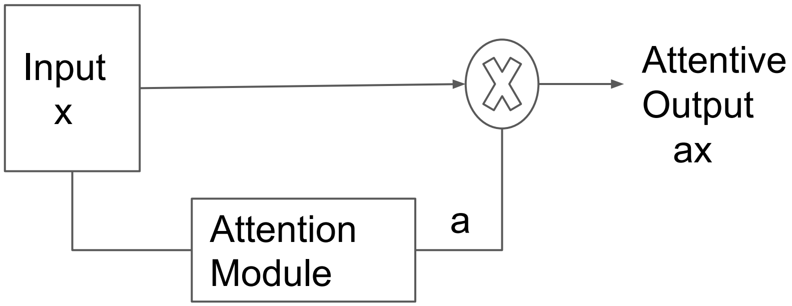
Reconstruction: Once trained, to "reconstruct" or generate an image, we start with pure noise at t=T.

- The model predicts the noise in the current image.
- We subtract a portion of that predicted noise to get a "slightly cleaner" image (t-1).
- We repeat this iteratively until we reach t=0

Self Attention in DNNs



Self Attention in DNNs



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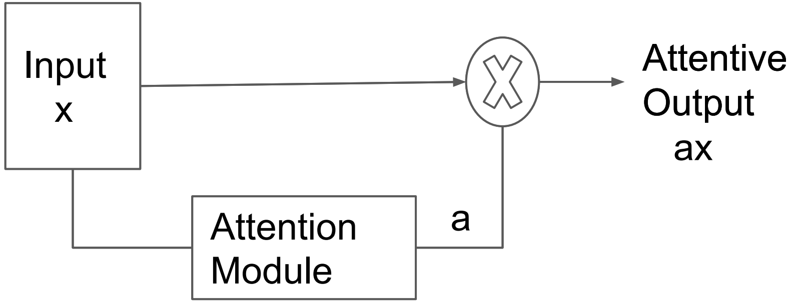
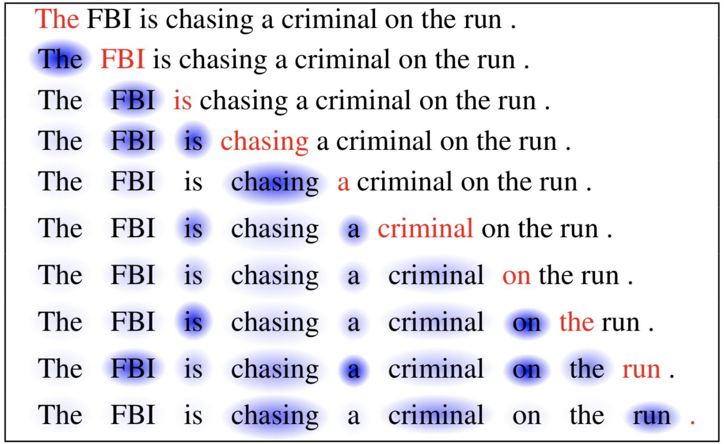
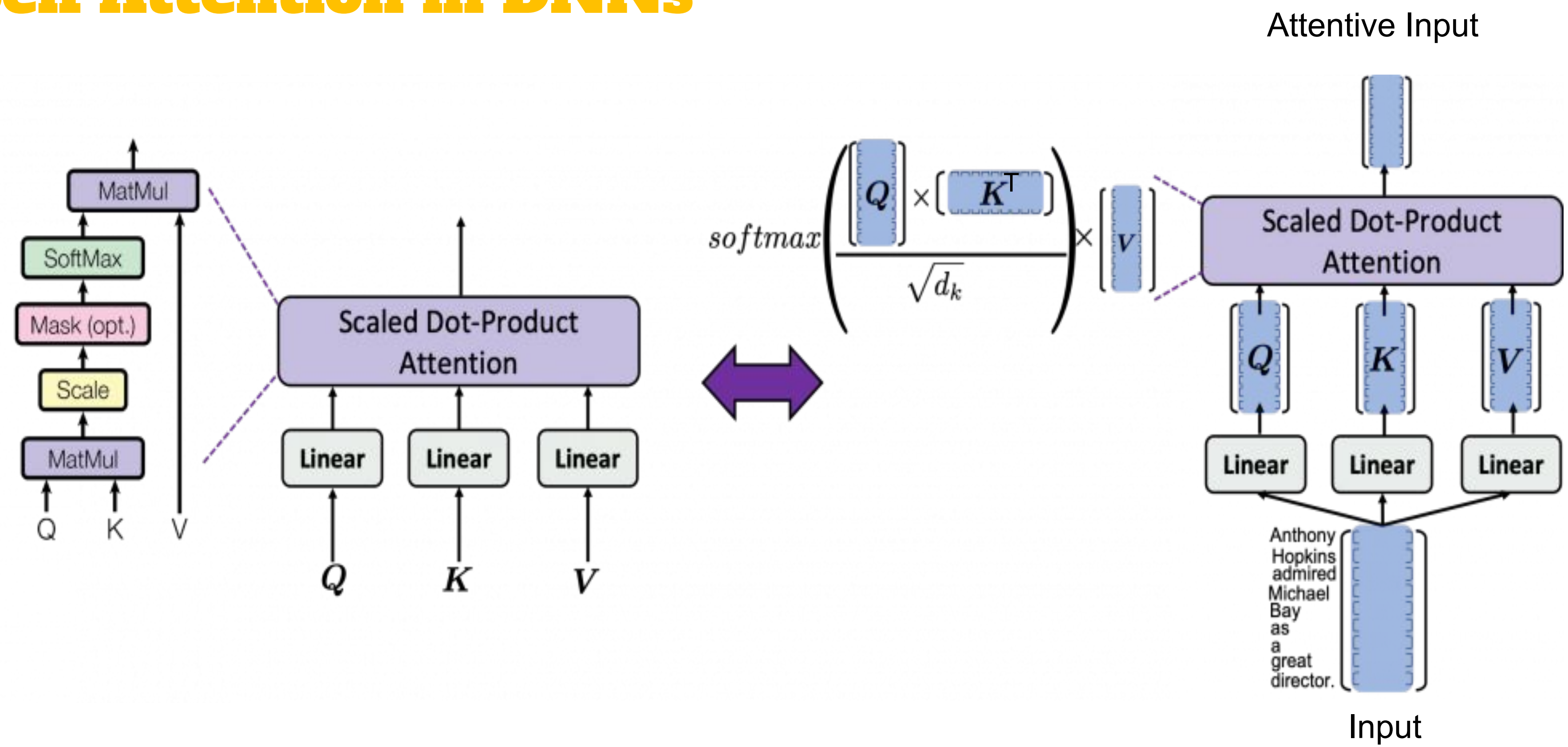
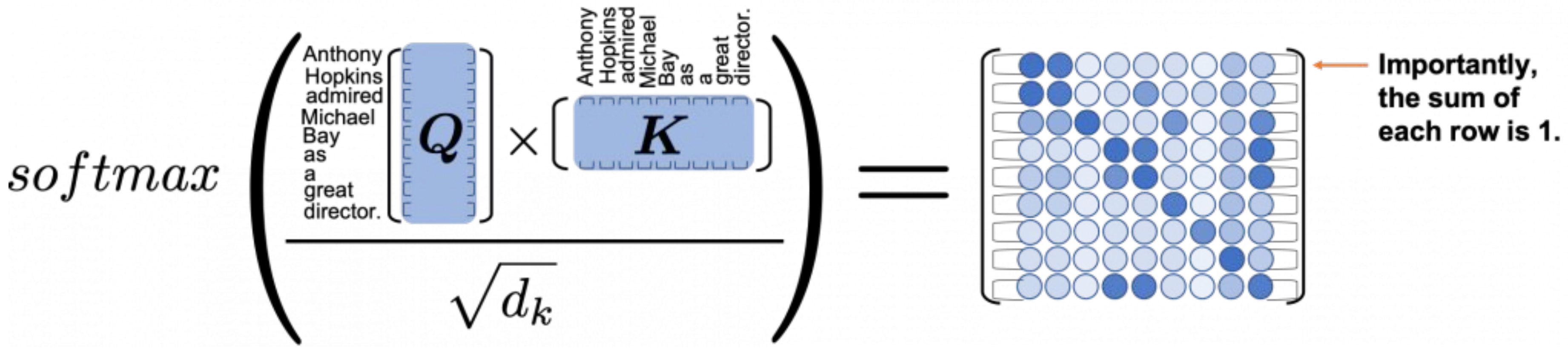


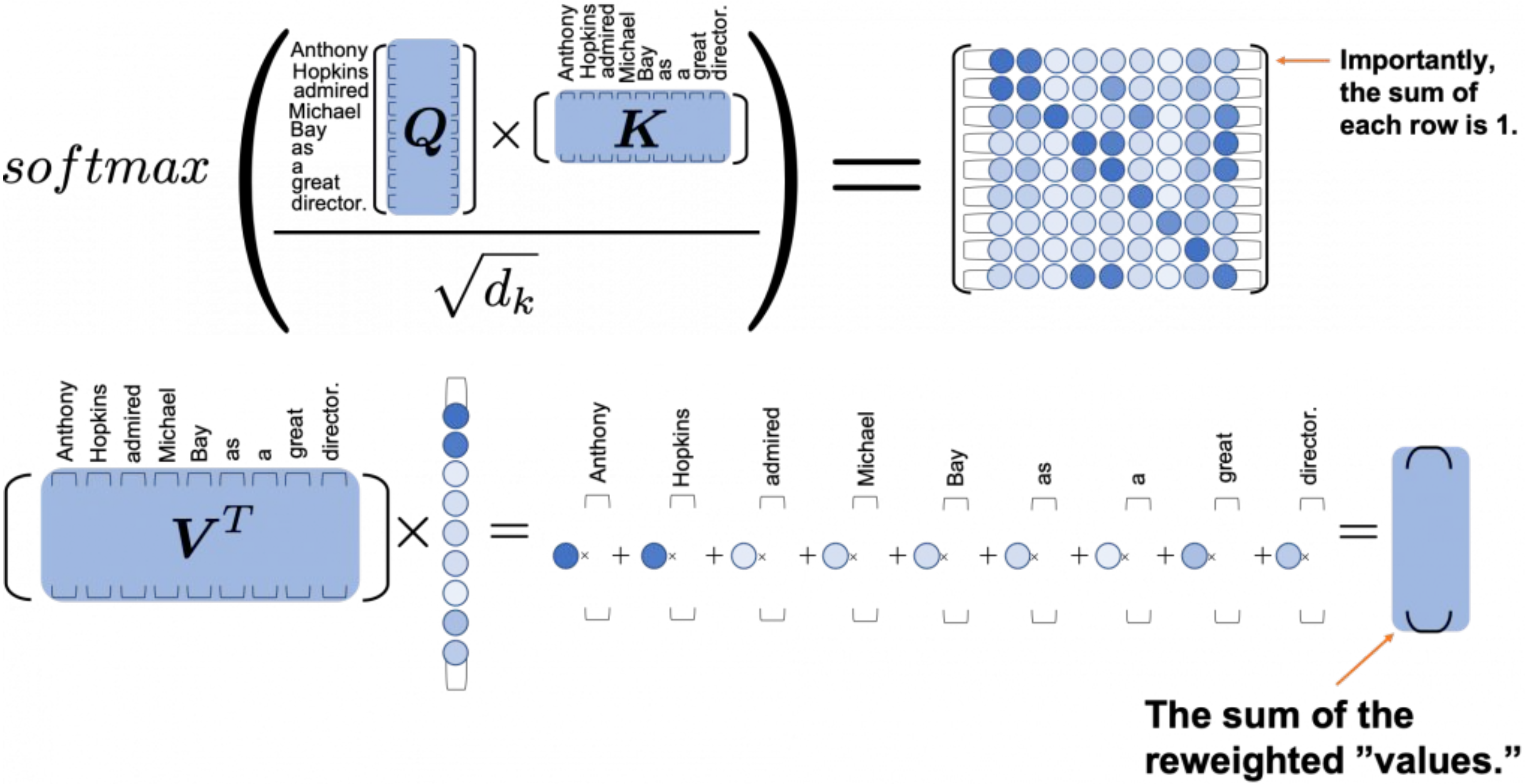


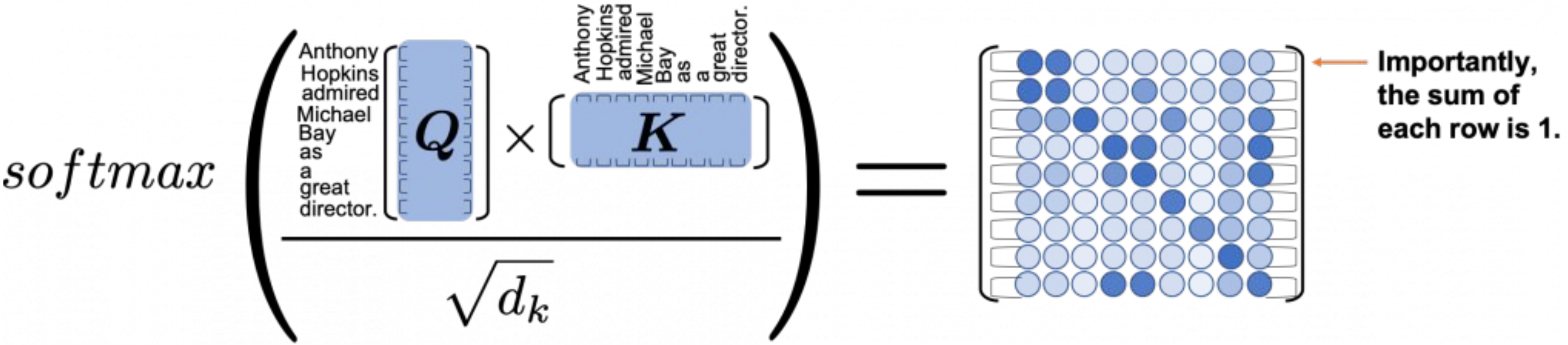
Self Attention in DNNs





Anthony Hopkins admired Michael Bay as a great director


softmax
Anthony Hopkins admired Michael Bay as a great director.
Q
K
$\sqrt{d_k}$
Importantly, the sum of each row is 1.
V^T
Anthony Hopkins admired Michael Bay as a great director.
The sum of the reweighted ”values.”



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Anthony
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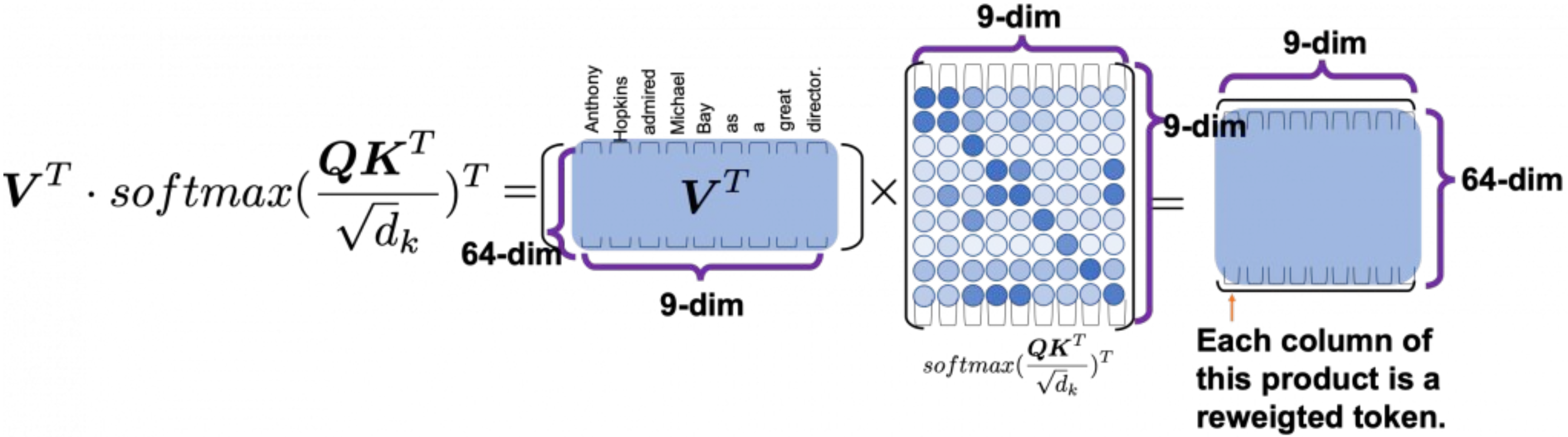
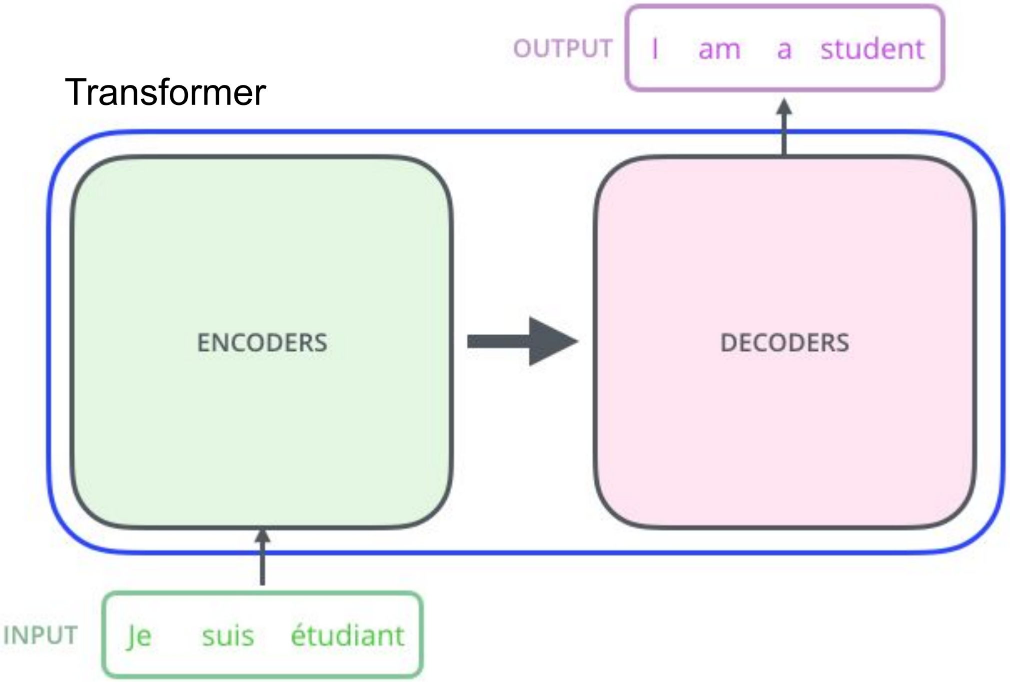
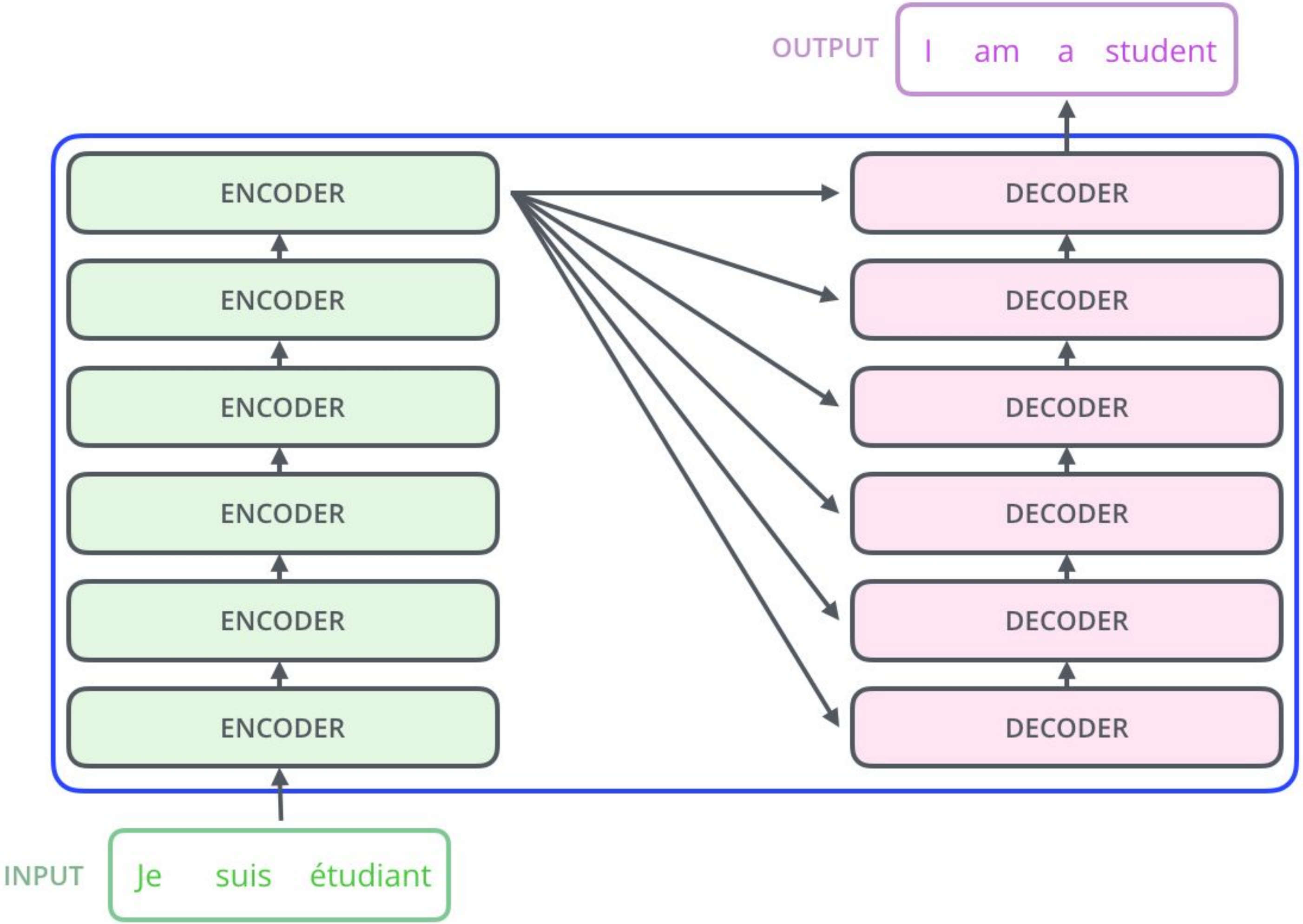


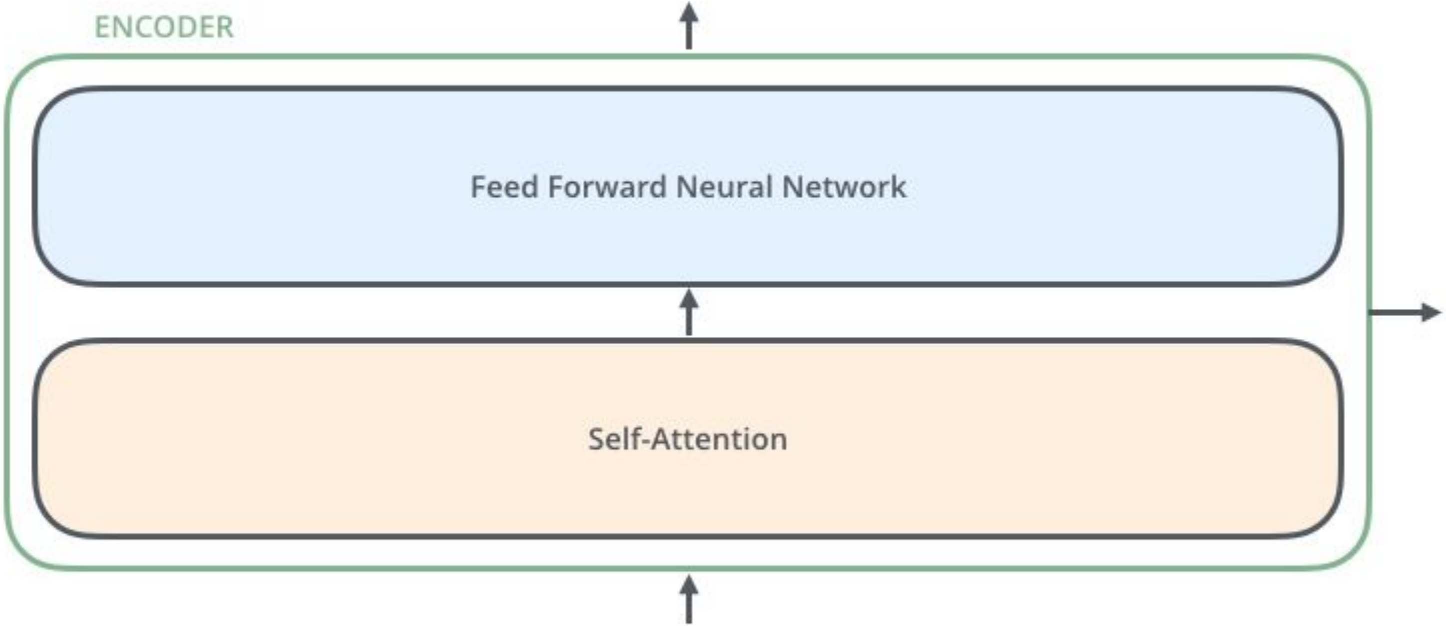
$V^T \cdot softmax(\frac{QK^T}{\sqrt{d_k}})^T =$
Anthony
Hopkins
admired
Michael
Bay
as
a
great
director.
V^T
64-dim
9-dim
9-dim
9-dim
$softmax(\frac{QK^T}{\sqrt{d_k}})^T$
9-dim
64-dim
Each column of this product is a reweigted token.

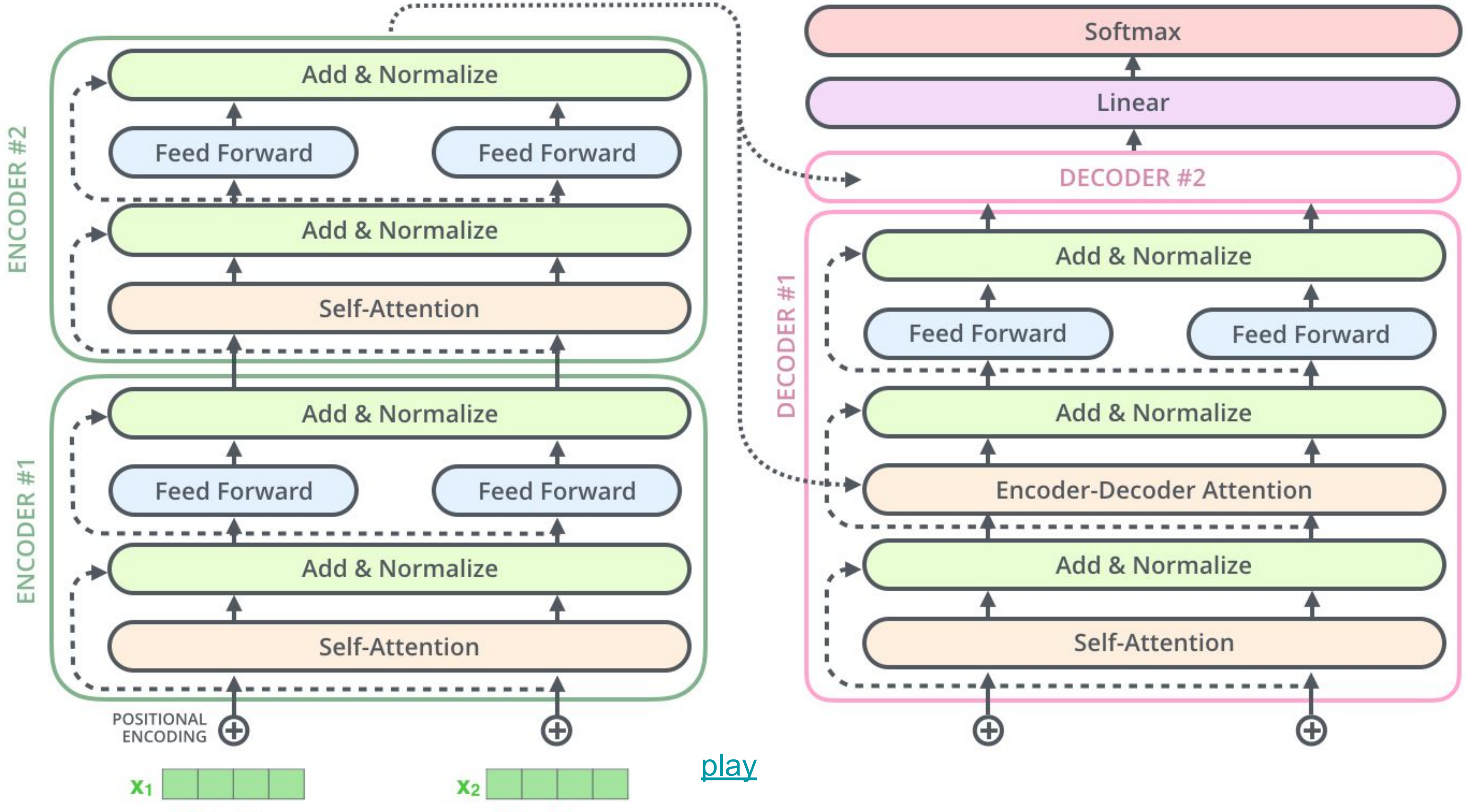
Transformer

Transformer



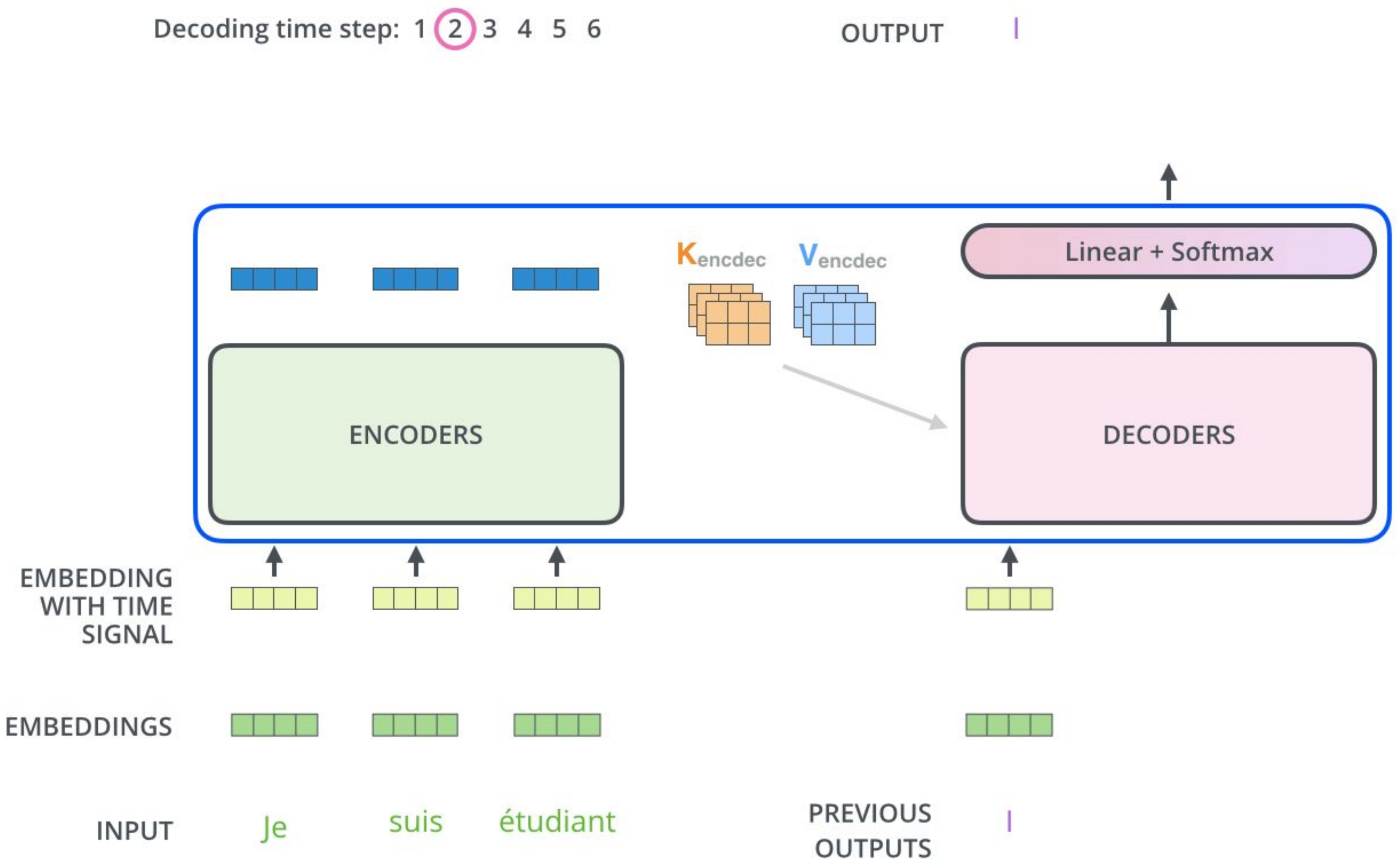

OUTPUT
I am a student
ENCODER
ENCODER
ENCODER
ENCODER
ENCODER
ENCODER
DECODER
DECODER
DECODER
DECODER
DECODER
DECODER
INPUT
Je suis étudiant


ENCODER
Feed Forward Neural Network
Self-Attention

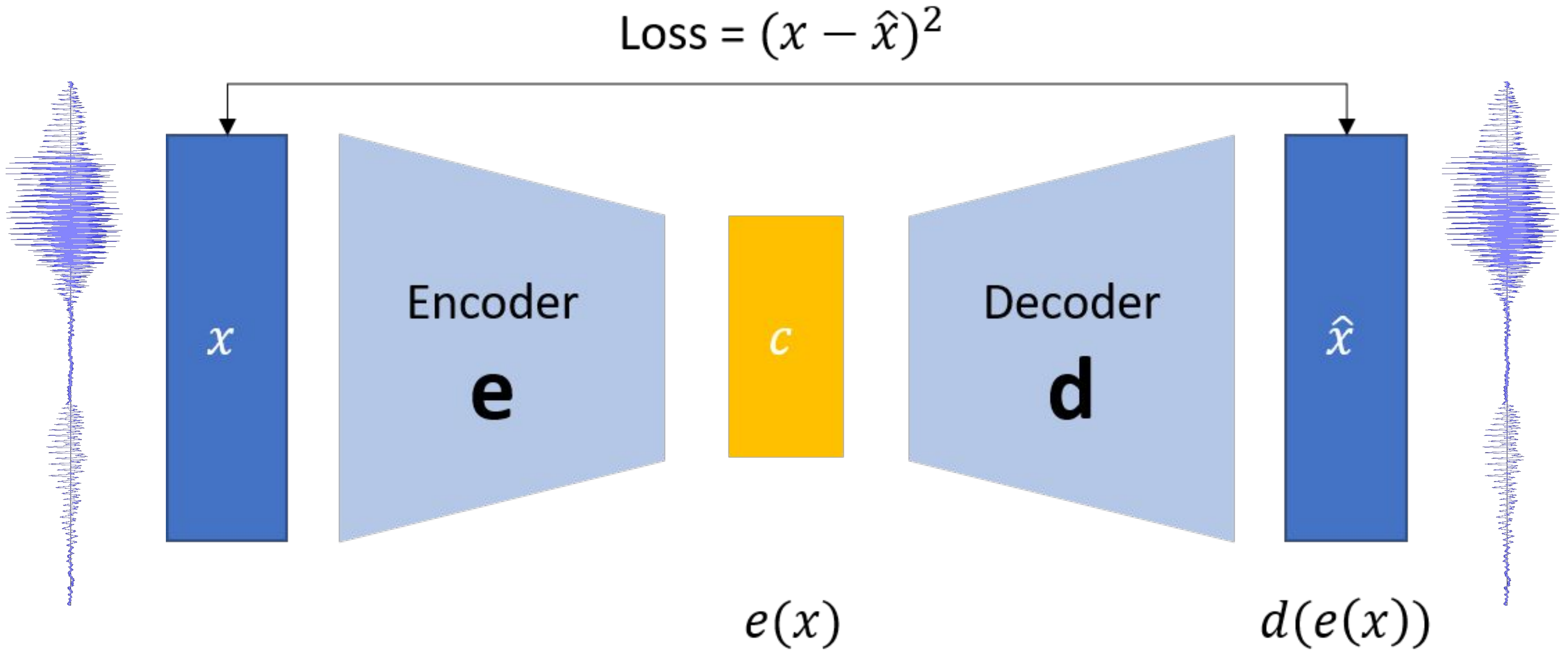


Softmax
Linear
DECODER #2
DECODER #1
Add & Normalize
Feed Forward
Feed Forward
Add & Normalize
Encoder-Decoder Attention
Add & Normalize
Self-Attention
ENCODER #2
Add & Normalize
Feed Forward
Feed Forward
Add & Normalize
Self-Attention
ENCODER #1
Add & Normalize
Feed Forward
Feed Forward
Add & Normalize
Self-Attention
POSITIONAL ENCODING
x_1
x_2

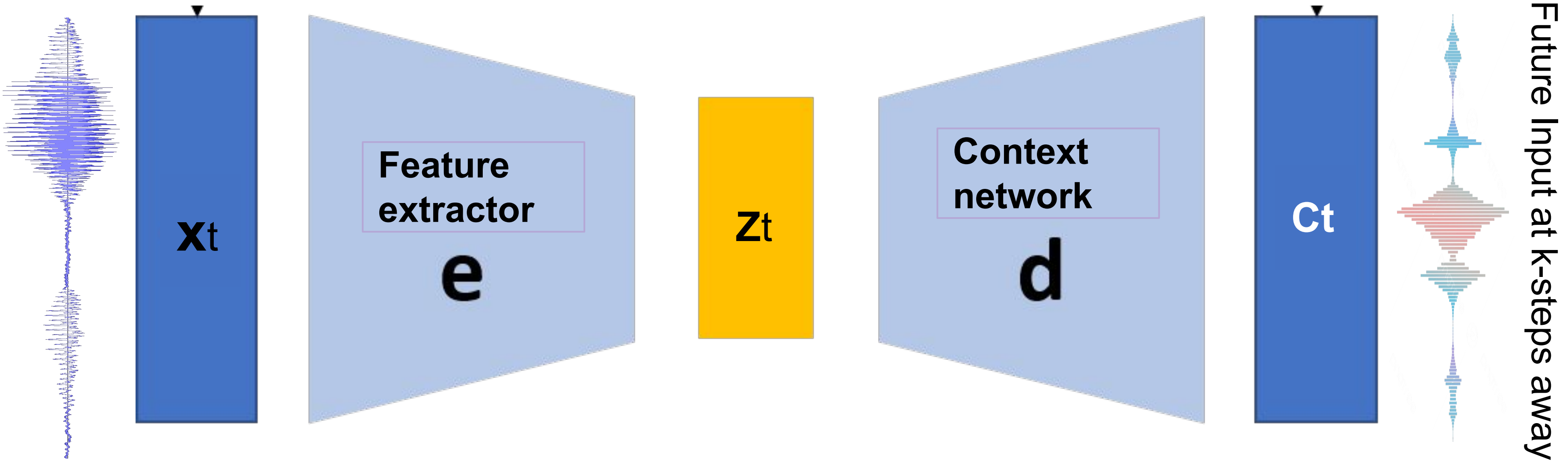
play


Decoding time step: 1 2 3 4 5 6
OUTPUT
I
Linear + Softmax
K_{encdec}
V_{encdec}
ENCODERS
DECODERS
EMBEDDING WITH TIME SIGNAL
EMBEDDINGS
INPUT
Je
suis
étudiant
PREVIOUS OUTPUTS
I

Autoencoder: Un-supervised Speech representations



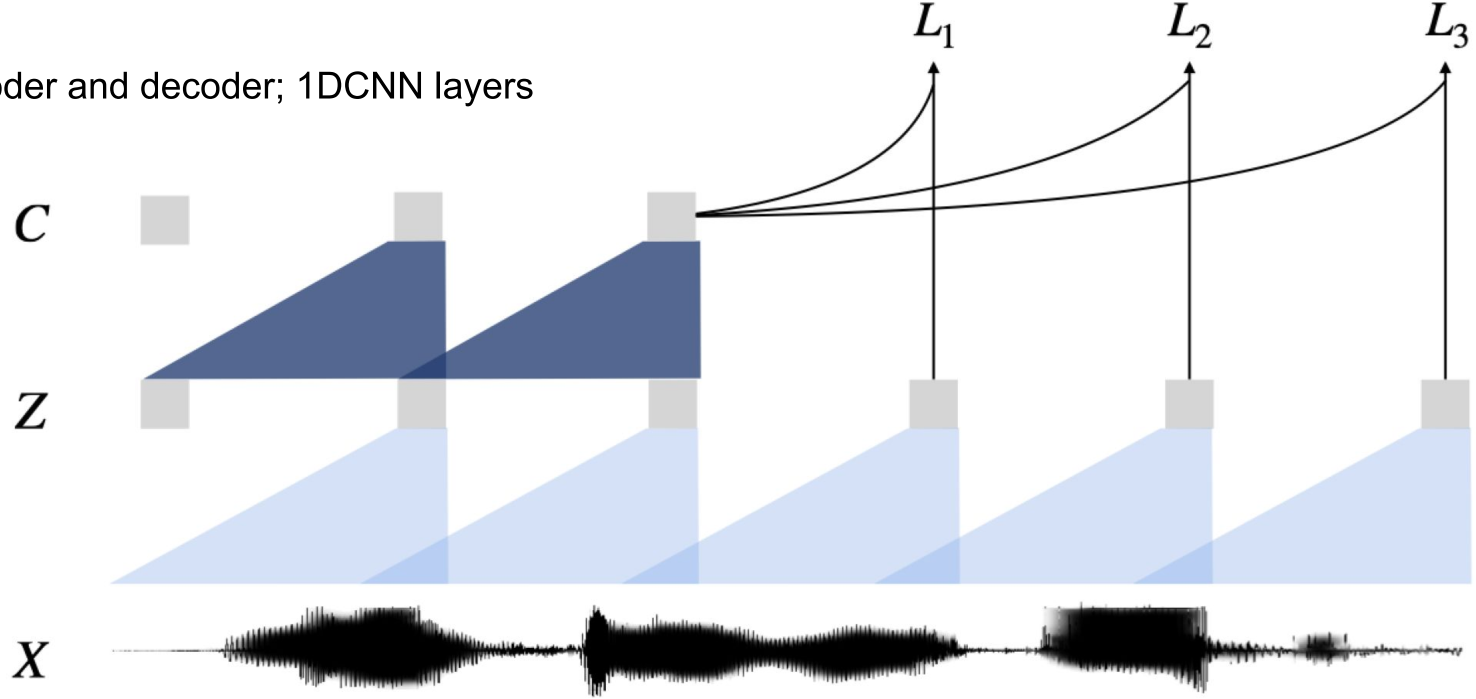
Wav2Vec: self-supervised Speech representations



Objective: Maximize mutual information between c_t and z_{t+k}

$$\mathcal{L}_k = -\sum_{i=1}^{T-k} \left(\log \sigma(\mathbf{z}_{i+k}^\top h_k(\mathbf{c}_i)) + \lambda \mathop{\mathbb{E}}_{\tilde{\mathbf{z}} \sim p_n} \left[\log \sigma(-\tilde{\mathbf{z}}^\top h_k(\mathbf{c}_i)) \right] \right)$$

Encoder and decoder; 1DCNN layers



Few Popular Architectures

- **ResNet:**
  - CNNs with skip Connections
  - Object classification
- **RCNNs:**
  - Region based CNNs
  - Face detection, Object tracking, Image Captioning
- **U-Net:**
  - CNNs as AutoEncoder
  - Image Segmentation, Remote sensing
- **Time Delay Neural Networks:**
  - 1D CNNs with time dilations
  - Speech Recognition, speaker ID
- **Siamese Network:**
  - RNNs/CNNs with Contrastive learning
  - Signature Verification